Behavioral Biometric-Based Continuous Authentication System for Secure Mobile Banking Using Machine Learning

Zeroth Review of CSD415 – Project Phase I

Department of Computer Science and Engineering
Government Engineering College Idukki

Team Members

Aiswarya R	IDK23CS006
Niranjana K D	IDK23CS045
Vimal Vijayan	IDK23CS063
Vishnu P S	IDK23CS064

Project Guide

Dr. Dhanya S Pankaj

Problem Statement

- **Objective:** To develop a behavioral biometric-based continuous authentication system for mobile banking applications, in order to verify user identity throughout an active session and prevent unauthorized access after login.

Motivation / Relevance of the Problem

Why is this problem important?

- Mobile banking apps handle highly sensitive financial data, making them prime targets once a session is hijacked.
- One-time, entry-point authentication cannot detect a change of user mid-session.
- Behavioral biometrics (typing pattern, touch, swipe, device motion) offer a passive, non-intrusive way to continuously confirm identity.

Expected Impact

- Reduces risk of unauthorized access after login (stolen/unlocked device scenarios).
- Maintains a smooth, frictionless user experience with no repeated manual verification.
- Enables real-time risk-based response via a dynamic trust score.
- Provides a scalable, ML-driven security layer that can integrate with existing banking apps via an API.

Existing Approaches and the Gap Identified

Existing Approaches

- Passive, multimodal authentication using touch, swipe, and sensor data [1].
- Siamese neural network approach for keystroke-dynamics-based verification [2].
- Anomaly-detection-based continuous authentication requiring only legitimate-user data [4].
- Behavioral authentication designed to generalize across usage scenarios [3].

Research Gap

- Most existing systems rely on a **single behavioral modality**.
- Many require **large labeled datasets** of both genuine and impostor behavior.
- Limited adaptability to **natural changes in user behavior** over time (behavioral drift).

Proposed Idea / Approach

- A **multimodal behavioral biometric framework** combining typing patterns, touch interactions, swipe gestures, and device motion to build individualized user profiles.
- **Anomaly detection algorithms** – One-Class SVM, Isolation Forest, and Autoencoders – will be comparatively evaluated to detect deviations from legitimate user behavior.
- A **dynamic trust score** is continuously generated to decide whether to:
  - Maintain the current session,
  - Request secondary authentication, or
  - Terminate access when suspicious activity is detected.
- A **mobile application** will be built to demonstrate the system in a live banking scenario, along with an **API** for integration with third-party platforms.

Feasibility and Expected Outcome

Feasibility

- Uses proven ML models (SVM, Isolation Forest, Autoencoders).
- Modular architecture enables independent development and testing.
- Public datasets and open-source tools ensure practical implementation.

Expected Outcome

- Continuous authentication system for mobile banking.
- Evaluated using FAR, FRR, EER, Precision, Recall, F1-Score, and latency.
- Mobile app prototype with integration-ready API.
- Scalable solution for behavioral biometric authentication.

References

1. G. Stragapede, R. Vera-Rodriguez, R. Tolosana, A. Morales, A. Acien, and G. Le Lan, “Mobile Behavioral Biometrics for Passive Authentication,” *Pattern Recognition Letters*, vol. 157, pp. 35–41, 2022.
2. A. Acien, A. Morales, J. V. Monaco, R. Vera-Rodriguez, and J. Fierrez, “TypeNet: Deep Learning Keystroke Biometrics,” *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 4, no. 1, pp. 57–70, Jan. 2022.
3. C. Lin, J. He, C. Shen, Q. Li, and Q. Wang, “CrossBehaAuth: Cross-Scenario Behavioral Biometrics Authentication Using Keystroke Dynamics,” *IEEE Transactions on Dependable and Secure Computing*, vol. 20, no. 3, pp. 2314–2327, May/June 2023.
4. M. Hu, D. Wang, C. Li, Y. Xu, and B. Tu, “Behavioral Biometrics-Based Continuous Authentication Using a Lightweight Latent Representation Masked One-Class Autoencoder,” *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 3, May/June 2025.